UCC 103 PRINCIPLES OF COMPUTING
Chapter 7 Impact of Computers in the Society

7.1 Introduction

Computing technologies, like most other forms of technology, are not socially neutral. They affect and are themselves affected by society. Computers have changed the way people relate to one another and their living environment, as well as how humans organize their work, their communities, and their time. Society, in turn, has influenced the development of computers through the needs people have for processing information. The study of these relationships has come to be known as "social informatics." Computing technology has evolved as a means of solving specific problems in human society.

The rapid rise of electronic computing in the 20th century changed the course of modern civilization. Although many of the effects have been positive, PCs have also impacted lives in undesirable ways. As computer technology continues to advance and new generations of machines grow faster and have greater capabilities, the machines become more deeply fixed in daily life, magnifying both the benefits and the downside risks.

7.2. Positive and Negative impacts

1) Easy Information Access: Computers have revolutionized the everyday access of information, whether for business, scientific or personal use. Computer software has become easy and intuitive to use, giving more people access to the information they want with less effort.
2) Automated Machinery**:** Computers now control many kinds of machines, vastly improving the convenience, safety and productivity of their use. For example, computers are crucial to the antilock brakes that are commonplace in vehicles. Motion sensors fitted to each wheel continually feed data to a dedicated computer.
3) Fast, Accurate Data Processing**:** In many areas, computers have taken on small, repetitive tasks, freeing people's time for more important endeavors. Before computers became fixtures in business, bookkeeping was handled by legions of clerks. Today, computers process millions of records in seconds, improving reliability and accuracy while reducing costs.
4) Sedentary Lifestyle**:** Computers have made video games, Web surfing and other forms of leisure possible. Although these pursuits led to the development of whole new industries, they also seduced many people into excessively sedentary habits. Time spent at the computer requires little physical exertion.
5) Family and Leisure Interruptions**:** Laptop computers and other mobile devices allow many people to work from home or be on call 24 hours a day, making it harder to keep work and home life separate. Work-related duties can interfere with family time at a moment's notice. The deterioration of boundaries between personal and professional time can strain relationships and make it difficult to relax when you're never completely away from the office.
6) Loss of Privacy: The widespread use of personal computers and mobile devices has led to an avalanche of personal data in electronic form, such as contact lists and Facebook posts. Before computers found a place in every home, much of this information was either disorganized or not connected. The Internet's ease of data gathering combined with deliberate efforts by businesses to exploit personal information has led to a widespread sharing of formerly private data.
7) Business transaction
   - E-commerce - shop online
   - Pay Bills
8) Communication
   - Email
   - Share information

- Chat platforms
- Web Conferencing

7.3. Impact on the Society

In this section we look at how computers have impacted on the various perspectives of the society.

7.3.1 At the National / Government Level

The impact of computers at the National /Government level is immense. For examples one third (1/3) of economic growth in the developed countries is attributed to digital technologies. Individual in the general population can make well informed decisions due to instant access to information from anywhere.

How the government use computers?

Government Web sites provide citizens with up-to-date information. They also allow citizens to file taxes, apply for permits and licenses, pay parking tickets etc.

7.3.2. Impact of Computers at Home and Work Place

Computer at home and workplace are used for: -

1) Entertainment
2) Research and education
3) Budgeting, personal financial management, and home/office business management
4) Web access
5) Personal and business communications
6) Video conferencing – holding meetings over the internet

The following are the main reasons computers have infiltrated homes and offices.

- Office work automation
- Web-based news, research
- Distance education
- Online shopping
- Download and listen to music
- Download and watch movies

Computers changed home/Office communications by transmitting text, voice, sounds, video, and graphics over the Web.

7.3.3. Impact of Computers in Education

There are various ways computers have impacted in education. Students have more tools to assist them in learning process. Some of these are outlined below: -

A. Computer-based training (CBT)
   1) Learn by using computers with instructional software
   2) Also called computer-aided instruction (CAI)
   3) Web-based training (WBT) uses Internet technology

   Advantages of CBT over traditional training are:-
   1) Self-paced study
   2) Unique content

3) Unique instructional experience, such as simulations
4) Reduced training costs

B. Distance learning (DL)
1) Delivery at one location and learning at another
2) Most colleges offer some form of distance learning; some include degrees

C. E electronic books (e-books)
Some Web sites allow you to download entire book to your computer. The Cost is about the same or less than cost of purchasing print version. It allows access of books not available in local book stores.

D. Computers in Edutainment
Edutainment is a type of educational software combining education with entertainment This is especially useful for early years of education and for learners with some challenges.

7.3.5. Social Impact of Computers

1) Digital divide in the society: Dividing the society between those who have the capacity and capability to use computers
2) Entertainment: Computers have created and availed multiple music and video options for entertainment as outlined below: -

    - Play CDs or DVDs
    - Online Audio and Video
    - Online Movies
    - Online Radio and Television
    - Purchase on Web and download to hard disk

3) Managing Finances: Computers are largely being use by individuals to balance their checkbooks, pay bills, track personal income and expenses, track investments, and evaluate financial plans
   They are also being used in online stock trading. This is buying and selling stocks online without using a broke. Individuals also use computers to organize other investments and set up alerts.

7.3.6. Impact of Computers in health care

Computers are used in health care to:-
1) Maintain and share patient records
2) Monitor patient's vital signs
3) Assist in research and diagnosis
4) File insurance claims
5) Conduct medical tests
6) Communicate with patients
7) Surgically implant computerized devices
8) Assist in operations with computer-controlled devices
9) Aid in surgery

Computer and the Web can aid Patients to:-
1) Provide up-to-date medical, fitness, nutrition, or exercise information

2) Access databases of doctors and dentists
3) Talk to others diagnosed with similar conditions using chat rooms
4) Order prescriptions Online

How can a computer aid in fitness?
Exercise equipment often has a built-in computer to

- track progress
- monitor physical conditions
  - heart rate
  - pulse

7.3.7. Impact of Computers at the Work Environment & How to ensure a Healthy Work Environment.

There are various areas of concern for a healthy work environment. Long-time computer use can lead to health Complication including; Job-related injuries and illnesses, Computer Addiction, Ergonomics and workplace design, and Green computing.
The following are the precautions to reduce the chance of developing tendonitis or carpal tunnel syndrome.

- Take frequent breaks during computer session
- Use wrist rest
- Exercise hands and arms
- Minimize number of times you switch between mouse and keyboard

What is computer addiction?
Occurs when computer consumes someone's entire social life

Symptoms
1) Craves computer time
2) Overjoyed when at computer
3) Unable to stop computer activity
4) Irritable when not at computer
5) Neglects family and friends
6) Problems at work or School

What is green computing?

- Involves reducing electricity and environmental wastes while using a computer

What are some green computing suggestions?

7.3.8. Impact of Computers in Science and Research

Scientists use computers for the following tasks: -
1) Help them collect, analyze, and model data
2) Use Internet to communicate with colleagues around world

An example of a scientific use of computer is the cochlear implant (chip) that allows deaf person to hear. The device does the following: -

- Microphone picks up sound and carries it to computer

- Computer digitizes sound
- Coil carries signals to implant
- Implant delivers electrical energy to electrodes in cochlea
- Electrodes stimulate auditory nerve fibers
- Auditory nerve sends sound information to brain.

7.4. Emerging Computer Technologies

There are numerous emerging computer technologies with varying impacts on society. We shall sample just a few.

7.4.1. Global Positioning System (GPS)

An example of use of this technology is in vehicles. The features of global positioning system (GPS) in a car are: -

1) Provide Directions
2) Make Reservations
3) Automatically call for help in emergency
4) Dispatch roadside Assistance
5) Track stolen Vehicle
6) Perform remote Diagnostics
7) Unlock door
8) Honk horn

7.4.2. E-Commerce

E-commerce electronic commerce or EC- is the buying and selling of goods and services, or the transmitting of funds or data, over an electronic network, primarily the internet. These business transactions occur either as business-to-business, business-to- consumer, consumer-to-consumer or consumer-to-business

For example, the following online e-commerce services are available for car buyers.

1) Shop online for a car
2) Obtain loans, leases, insurance, and warranties
3) Sell a used Car

E- commerce facilitates the concept of telecommute. Telecommute has the following advantages: -

Reduce time and expense spent in traveling to office
- Eliminate travel during unsafe weather conditions
- Allow flexible work schedule
- Provide convenient, comfortable work environment
- Further advantages

– Higher job satisfaction rates
– More productive workers
– Reduction in employer overhead
– Healthy for environment

7.4.3 Social Media

1) Faceboook
2) Twitter

7.4.4. Chat Forums

7.5. Computer Ethics

Computer ethics is defined as the analysis of the nature and social impact of **computer** technology and the corresponding formulation and justification of policies for the **ethical** use of such technology.

- **Computer ethics** are a set of moral standards that govern the use of computers.

7.5.1. Computer Ethics Issues

Ethics deals with good, right and just behaviour as well as evil, wrong and unjust behaviour. It is important because it can affect a stakeholder's ability to achieve his/her
goals. Areas of computer ethics include the following: -

1) Unauthorized use of computer systems
2) Information privacy
   Privacy Concerns;
   a) **Hacking** – is unlawful intrusion into a computer or a network. A hacker can intrude through the security levels of a computer system or network and can acquire unauthorised access to other computers.
   b) **Malware** – means malicious software which is created to impair a computer system. Common malware are viruses, spyware, worms and trojan horses. A virus can delete files from a hard drive while a spyware can collect data from a computer.
   c) **Data Protection** – also known as information privacy or data privacy is the process of safeguarding data which intends to influence a balance between individual privacy rights while still authorising data to be used for business purposes.
   d) **Anonymity** – is a way of keeping a user's identity masked through various applications.
3) Software theft (piracy)
4) Information Accuracy
5) Intellectual property rights
   a) **Copyright** – is a form of intellectual property that gives proprietary publication, distribution and usage rights for the author. This means that whatever idea the author created cannot be employed or disseminated by anyone else without the permission of the author.
   b) **Plagiarism** – is an act of copying and publishing another person's work without proper citation. It's like stealing someone else's work and releasing it as your own work.
   c) **Cracking** – is a way of breaking into a system by getting past the security features of the system. It's a way of skipping the registration and authentication steps when installing a software.
   d) **Software License** – allows the use of digital material by following the license agreement. Ownership remains with the original copyright owner, users are just granted licenses to use the material based on the agreement.
6) Codes of conduct: A **code of conduct** is a set of rules outlining the norms, rules, and responsibilities or proper practices of an individual party or an organization.
7) Effects on Society
   a) **Jobs** – Some jobs have been abolished while some jobs have become simpler as computers have taken over companies and businesses. Things can now be done in just one click whereas before it takes multiple steps to perform a task. This change may be considered unethical as it limits the skills of the employees.
   b) Employees health; There are also ethical concerns on health and safety of employees getting sick from constant sitting, staring at computer screens and typing on the keyboard or clicking on the mouse.

c) **Environmental Impact** – Environment has been affected by computers and the internet since so much time spent using computers increases energy usage which in turn increases the emission of greenhouse gases.
    - Electronic Waste issues

d) **Social Impact** – Computers and the internet help people stay in touch with family and friends. Social media has been very popular nowadays.
    - Computer gaming influenced society both positively and negatively. Positive effects are improved hand-eye coordination, stress relief and improved strategic thinking. Negative effects are addiction of gamers, isolation from the real world and exposure to violence.
    - Computer technology helps the government in improving services to its citizens. Advanced database can hold huge data being collected and analysed by the government.
    - Computer technology aids businesses by automating processes, reports and analysis.